Variable

A variable is a container that stores data in a program. we can use the variable name to access.

In [ ]:
```
#Creating variables
a=10 # variable = value
```

In [ ]:
```
#string
name = "Reeta"
print(name)
print(type(name))
```

In [ ]:
```
#integer
age = 20
print(age)
print(type(age))
```

In [ ]:
```
#float
fee=20000.00
print(fee)
print(type(fee))
```

In [ ]:
```
#boolean
result = False
print(result)
print(type(result))
```

In [ ]:
```
#changing variabl value
age = 20
age = 21
print(age)
```

In [ ]:
```
#multiple variable
name,age = "Reeta",20
print(name)
print(age)
```

In [ ]:
```
#assign same value
a=b=c=10
print(a)
print(b)
print(c)
```

Casting = Casting means converting one data type to another data type.

In [ ]:
```
# str to int
age="20"
age=int(age)
print(type(age))
#boolean all num comes True(postive , negative)expect 0 and Flase(zero).
```

In [14]:
```
# float to int
CGPA = 8.51
per =int(CGPA)
print(per)
```

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In [12]:
```
#boolean to int
Grade = True
rank = int(Grade)
print(rank)
```

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Naming Conventions

Rules for naming variables, functions, and other objects in Python.

Types:

1. Camel Case
2. Pascal Case
3. Snake Case

In [19]:
```
#1. Camel Case: First word starts with a small letter; other words start with capital letters.
myName = "Reeta"
print(myName)
```

Reeta

In [23]:
```
#2. Pascal Case: Every word starts with a capital letter.
MyName = "Reeta"
print(MyName)
```

Reeta

In [25]:
```
#3. Snake Case: Words are separated by underscores (_).
my_name = "Reeta"
print(my_name)
```

Reeta

Variable Scope

Local variable: Created inside a function.

Global variable: Created outside a function.

In [44]:
```
num1 = 10 #global
def sum():
    num2 = 20 #local
    print(num1 + num2)
sum()
```

30

In [31]:
```
#Variable Reassignment
a = 20
a = 22
print(a)
```

22

In [27]:
```
#Deleting Variables (del)
num1 = 10
num2 = 20
del num1
print(num1)
print(num2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[27], line 5
      3 num2 = 20
      4 del num1
----> 5 print(num1)
      6 print(num2)

NameError: name 'num1' is not defined
```

In [29]:
```
num1 = 10
num2 = 20
del num1
print(num2)
```

20

In [52]:
```
#Object Identity (id()) - The id() function returns an object's identity.
a = 10
b = 20
print(id(a))
print(id(b))
```

```
140724664543960
140724664544280
```

In [54]:
```
#Unpacking Variables
#Assigning multiple values to multiple variables in one line
a, b, c = 10, 20, 30

print(a)
print(b)
print(c)
```

```
10
20
30
```

In [ ]: